



Decision Structures in Python

Practical Skills Module 3: Topic 1

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Topic Code	PM-03-PS01
Topic	Create decisions in Python

Scope of Practical Skill

Given a range the learner must be able to:

- PA0101 Create decisions in Python using IF statement.
- PA0102 Create decisions in Python using IF...ELSE statement.
- PA0103 Use ELIF to check for multiple expressions.
- PA0104 Create a FOR LOOP in Python
- PA0105 Create a WHILE LOOP in Python
- PA0106 Alter the flow of a loop using break and continue.
- PA0107 Use a pass statement inside a function

Applied Knowledge

- AK0101 Concept, definition and functions of Python programming functionalities

Internal Assessment Criteria

- IAC0101 Expected results with Python loops are achieved

Resources:

Knowledge	Information from Python Programmer Curriculum
National Curriculum Framework	900221-000-00-PM-03



Practical task

Follow the facilitator's instructions to complete the following activities:

Create a Python file named **inventory_manager.py** to implement the following tasks and ensure it is stored in your task folder.

Scenario: Inventory Management in a Warehouse

A warehouse manager is looking to automate inventory tracking and decision-making processes. Your goal is to build a basic inventory management system in Python that performs essential tasks such as checking stock levels, identifying low or overstocked items, processing restocks, and preparing for future alert features.

Your implementation should demonstrate key Python control structures while performing practical inventory operations.

Step 1: Identify Low Stock Using an if Statement

Write code that checks whether a specific item is low in stock (e.g. stock quantity less than 10). If so, mark the item for restocking.

Step 2: Evaluate Stock Levels Using an if...else Statement

Update your logic to handle two outcomes:

- If the item is low in stock, mark it for restocking.
- Otherwise, indicate that the stock level is sufficient.

Step 3: Use if...elif...else to Categorise Stock Status

Extend your decision-making to categorise stock into three levels:

- Low stock: less than 10 units
- Sufficient stock: between 10 and 30 units
- Overstocked: more than 30 units
Print a message for each status accordingly.

Step 4: Iterate Over Items Using a for Loop

Create a dictionary or list containing the following items and their quantities:

- Item A: 10
- Item B: 20

- Item C: 5

Use a for loop to go through each item and print its name along with its stock status (using the logic from Step 3).

Step 5: Monitor Restocking with a while Loop

Simulate restocking an item until it reaches a target stock level of 30. Print the stock level at each step to show the restocking process.

Step 6: Control Loop Execution with break and continue

Enhance your restocking logic to include:

- A maximum stock limit of 50. Use break to stop the restocking process if this limit is exceeded.
- If an item is already at or above the target stock, use continue to skip restocking it.

Step 7: Prepare for Future Features with pass

Define a placeholder function named `send_alert(item_name)` that will eventually be used to send alerts when stock is low. Use the pass statement inside the function for now.



Take note

Important: Be sure to upload all files required for the task submission inside your task folder and then click "Request review" on your dashboard.
